

Mini Project VII Report

MedScan Diagnostics - Blood Cell Detection with Yolo26

1. Introduction

1.1 Problem Statement

Manual analysis of blood smear images is a time-intensive diagnostic task that requires trained pathologists to identify and count individual blood cells under a microscope. In a clinical setting, accurate cell counts are critical for diagnosing a wide range of conditions, from anemia (abnormal RBC counts) to infection and leukemia (abnormal WBC counts) to thrombocytopenia (low Platelet counts). Manual counting is subject to human fatigue, inter-observer variability, and throughput limitations, particularly in high-volume diagnostic labs.

This project investigates whether a modern real-time object detector, YOLO26 (Ultralytics), can automate the detection and classification of the three primary blood cell types: Red Blood Cells (RBC), White Blood Cells (WBC), and Platelets, from microscopy images with sufficient accuracy to serve as a clinical screening aid.

1.2 Business Motivation

For MedScan Diagnostics, automating blood cell detection offers several potential benefits:

- Reduced pathologist workload by pre-screening routine samples automatically.
- Faster turnaround times for standard blood count tests.
- Consistent, fatigue-free performance across high sample volumes.
- Flagging of abnormal samples for priority human review.

The target application is a screening assistant, not a diagnostic replacement. A licensed pathologist remains the final decision-maker for any clinical finding.

2 Methodology

2.1 Dataset

The BCCD (Blood Cell Count and Detection) dataset was sourced from Roboflow Universe (Joseph Nelson, Version 3, MIT License). It contains annotated microscopy images of blood smears split as follows:

- Train: 255 images
- Validation: 73 images
- Test: 36 images

The training set contains 3,450 total annotated instances across three classes: Platelet (249), RBC (2,938), and WBC (263). This creates a significant class imbalance, with RBC representing approximately 85% of all instances.

2.2 Model Selection

YOLO26n (Nano) was selected as the detection model. YOLO26 is the latest generation of the Ultralytics YOLO family, offering improved accuracy over prior versions while maintaining real-time inference speeds. The Nano variant (2.37 million to 2.57 million parameters) was chosen for its balance of speed and accuracy, appropriate for a classroom training environment with GPU acceleration (NVIDIA GeForce RTX 3060, 12GB VRAM). All runs were initialized from pre-trained COCO weights (yolo26n.pt), enabling transfer learning from a large general-purpose detection dataset.

2.3 Hyperparameter Experiments

Four configurations were evaluated to identify the optimal batch size and input resolution combination. All other hyperparameters were held constant: 25 epochs, AdamW optimizer, COCO pre-trained weights, confidence threshold 0.25, IoU threshold 0.45.

Run	Batch	Img Size	mAP@50	mAP@50-95	Notes
A — batch16 / imgsz416	16	416	0.8622	0.6082	Smaller input, faster
B — batch16 / imgsz640	16	640	0.8671	0.6270	Best overall
C — batch8 / imgsz416	8	416	0.8542	0.6109	Smaller batch
D — batch8 / imgsz640	8	640	0.8541	0.6092	Large input, small batch

The best over model was B - batch 16 with image size 640.

2.4 Training Setup

Training was conducted in a Jupyter notebook environment on a local GPU workstation. Key settings common to all runs:

- Architecture: YOLO26n (Nano), pre-trained on COCO
- Epochs: 25 (with early stopping patience = 10)
- Optimizer: AdamW (auto-selected by Ultralytics)
- Data augmentation: Ultralytics defaults (mosaic, HSV jitter, random flip, scale)
- Validation: Roboflow validation split (73 images) after each epoch

3. Results

3.1 Hyperparameter Comparison: Image Size Effect

Comparing runs with the same batch size, larger image resolution (640px) consistently outperformed 416px:

- Batch 16: imgsz 640 scored mAP@50 = 0.8671 vs. imgsz 416 at 0.8622 — a gain of +0.0049.
- Batch 8: imgsz 640 scored mAP@50 = 0.8541 vs. imgsz 416 at 0.8542 — essentially tied, with 640 slightly ahead on mAP@50-95 (0.6092 vs. 0.6109 favours 416 marginally).

The improvement from 640px is most meaningful for small objects. Platelets are particularly small relative to the image field, and the higher resolution provides more spatial context for the model to distinguish them from background debris. The AP@50 for Platelets in Run B (0.8168) was slightly higher than in Run A (0.7881), supporting this interpretation.

3.1.2 Hyperparameter Comparison: Batch Size Effect

Comparing runs with the same image size, batch size 16 consistently outperformed batch size 8:

- imgsz 416: Batch 16 scored 0.8622 vs. Batch 8 at 0.8542 — a gap of +0.0080.
- imgsz 640: Batch 16 scored 0.8671 vs. Batch 8 at 0.8541 — a larger gap of +0.0130.

With only 255 training images, batch size 16 provides more stable gradient estimates per update step compared to batch size 8. Smaller batches introduce higher gradient variance, which can make training noisier and slower to converge, especially on a small dataset. The precision scores reflect this: Batch 16 runs achieved precision of 0.7779 and 0.8195, while Batch 8 runs scored 0.7570 and 0.7839 respectively — consistently lower, suggesting the model was less certain about its positive detections under smaller batch training.

3.2 Best Model — Overall Metrics

The best model (Run B) achieved the following on the validation set:

Metric	Value
mAP@50	0.8671
mAP@50-95	0.6270
Precision	0.8195
Recall	0.8253

3.3 Per-Class Performance

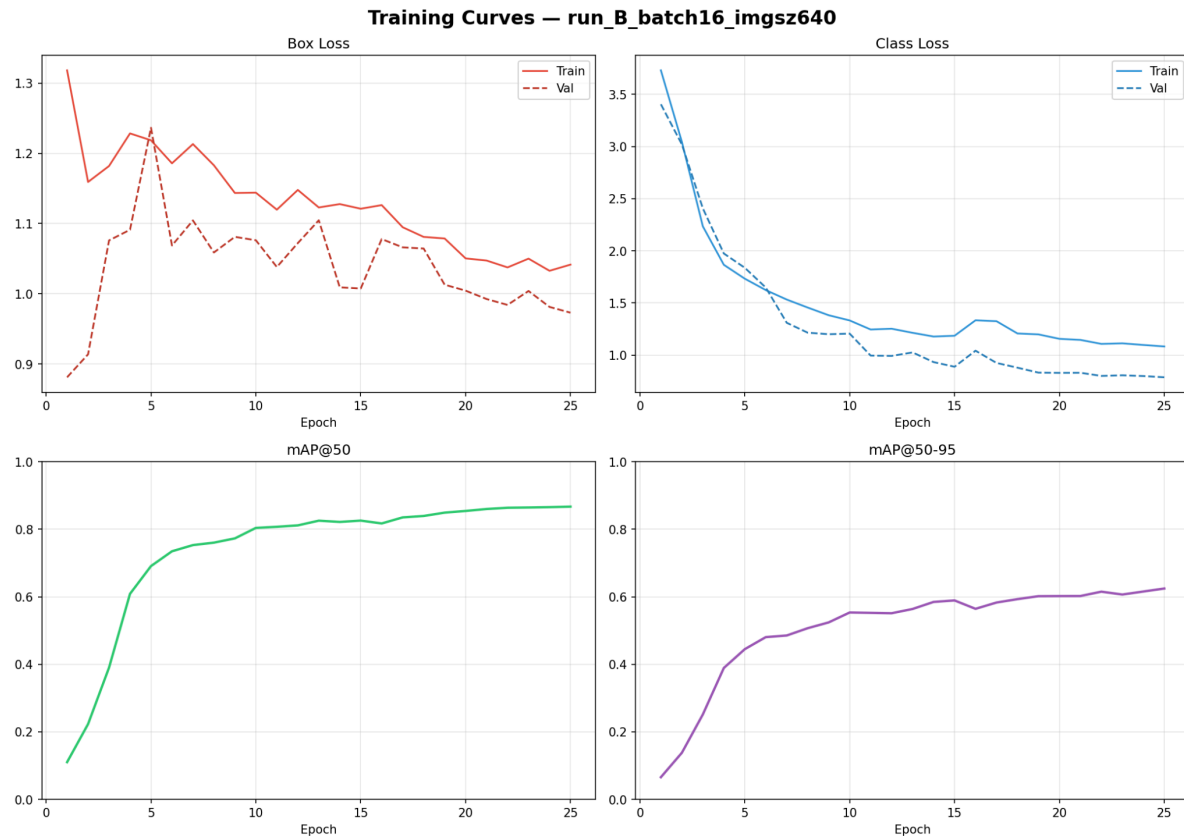
Class	AP@50	AP@50-95	Notes
Platelet	0.8168	0.4959	Small object; limited training examples (249)
RBC	0.8057	0.5795	Dense overlap — model's weakest class
WBC	0.9787	0.8057	Visually distinctive; near-perfect AP
mAP (mean)	0.8671	0.6270	Run B — best config

WBC detection was near-perfect at $AP@50 = 0.9787$. This reflects the class's strong visual distinctiveness: WBCs are large, darkly stained, and well-separated from surrounding cells, making them easy to localize even from limited training examples (263 instances). RBC achieved the lowest $AP@50$ at 0.8057, not due to data scarcity (2,938 instances — 85% of the training set) but because RBCs are densely packed and visually similar to each other, making precise individual localization difficult. Platelet performance (0.8168) is limited by two factors: the cells are physically small and the model had only 249 training examples.

3.4 Training Curves – Best Model(Run B)

Both box loss and class loss dropped sharply in the first 5 epochs and continued declining gradually through epoch 25, with validation loss tracking slightly below training loss throughout, which is a healthy sign of good generalization without overfitting. The $mAP@50$ rose steeply from near zero to approximately 0.80 by epoch 7, then improved more gradually, reaching 0.8671 by epoch 25.

The $mAP@50-95$ showed a similar trajectory, climbing to 0.627 by the end of training. Neither metric had fully plateaued at epoch 25, suggesting that additional epochs would likely yield further improvement. Given the early stopping patience of 10 epochs was not triggered, the model was still learning at the end of training.

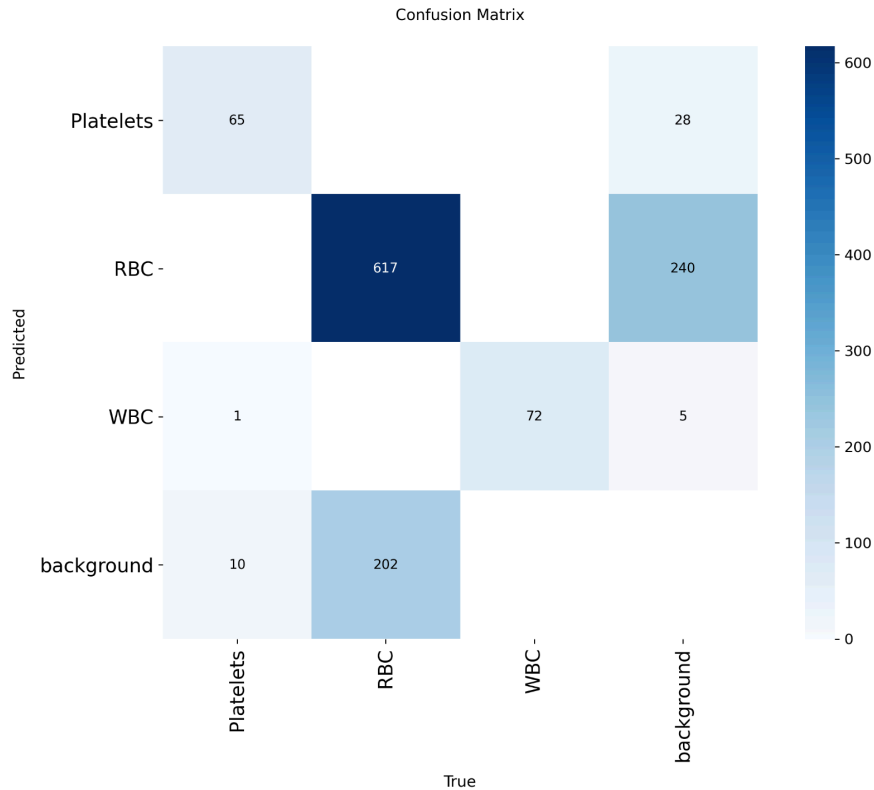


3.5 Confusion Matrix

The confusion matrix for the best model reveals the following key patterns:

- 65 of 75 Platelets correctly detected; 10 missed as background, with 28 false positives from background regions.
- 617 of 819 RBCs correctly detected; 202 missed as background, with 240 false positives; the model's weakest area.
- 72 of 72 WBCs correctly detected with only 5 false positives; near-perfect performance.

The dominant error mode is RBC-to-background confusion in both directions. This is consistent with the dense, overlapping nature of RBCs in blood smear images, where closely packed cells make it difficult for the model to distinguish individual cell boundaries from cluttered background regions.

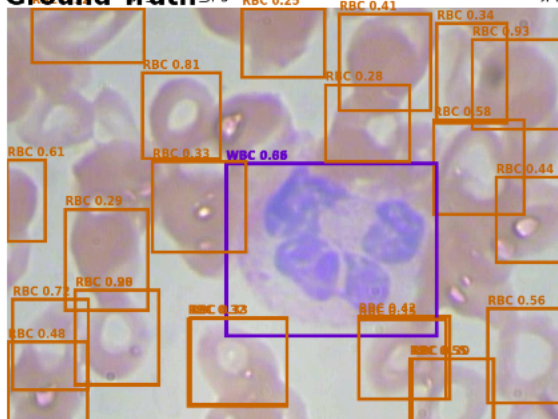
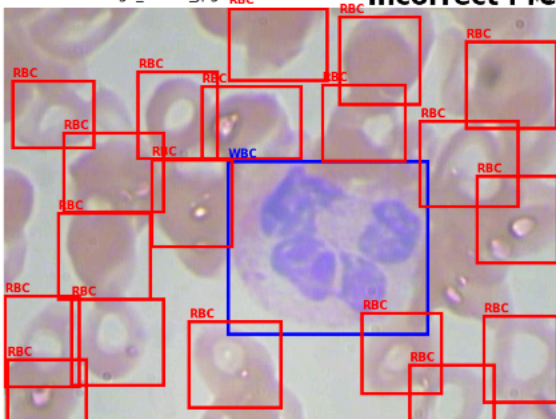


3.6 Incorrect Predictions – Error Analysis

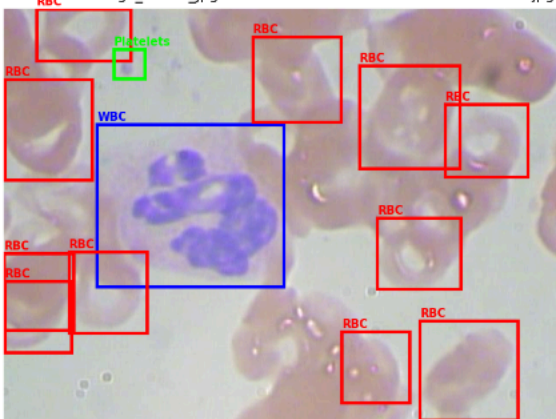
The six validation images above show cases where the model's predictions diverged from ground truth. WBC detection remains confident and accurate across all examples (confidence 0.79–0.99), consistent with its near-perfect AP score. Platelet detection is mostly correct but noticeably less confident (0.41–0.60), with some instances missed entirely, reflecting the challenge of detecting small objects at this resolution.

RBC is where the most errors occur. Comparing ground truth (left) to predictions (right), several labeled RBCs have no corresponding prediction, particularly for cells that are partially occluded or packed tightly at the edges of clusters. At the same time, the model generates extra RBC detections in unlabeled background regions, producing false positives. Many surviving RBC predictions carry low confidence scores (0.27–0.45), indicating the model is uncertain about partially visible or heavily overlapping cells. These patterns are consistent with the confusion matrix results and confirm that dense cell overlap is the primary challenge for this model on the BCCD dataset.

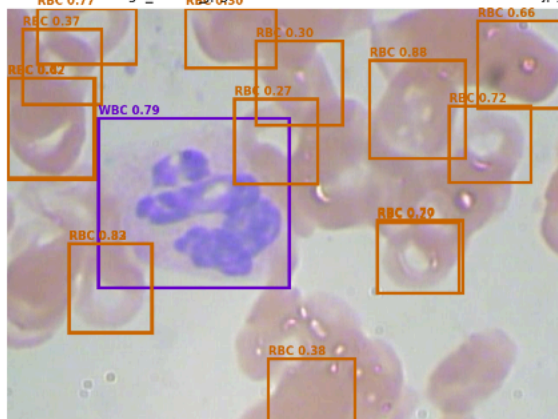
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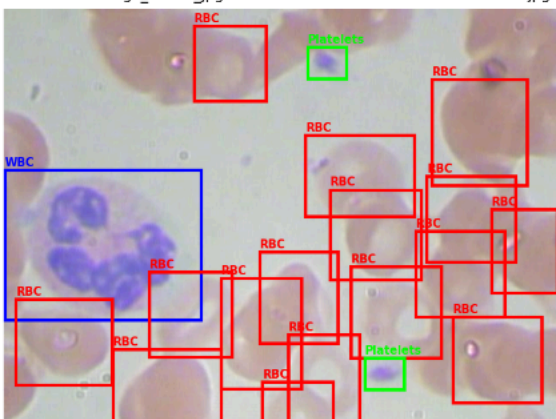
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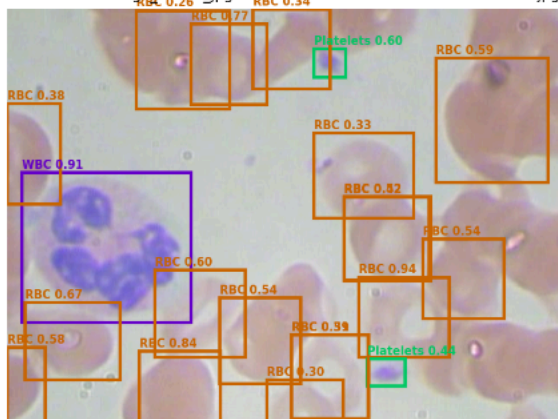
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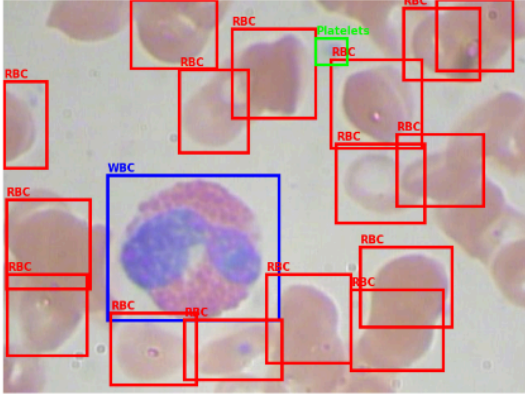
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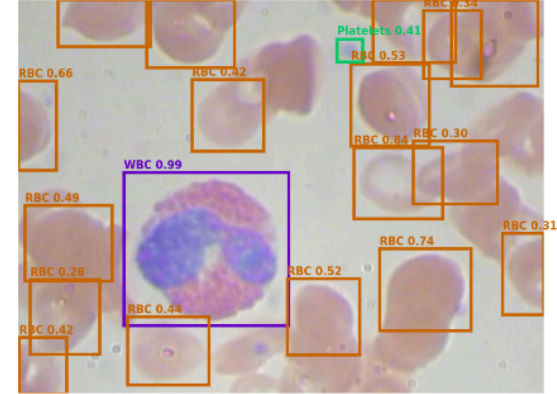
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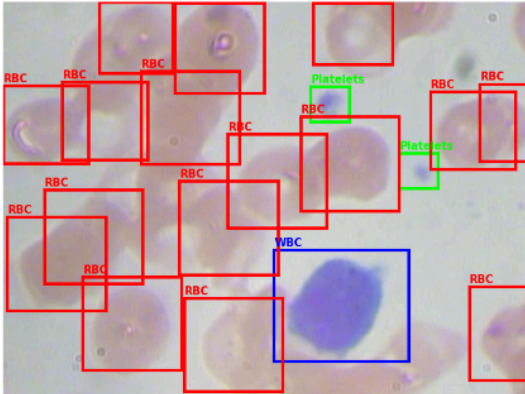
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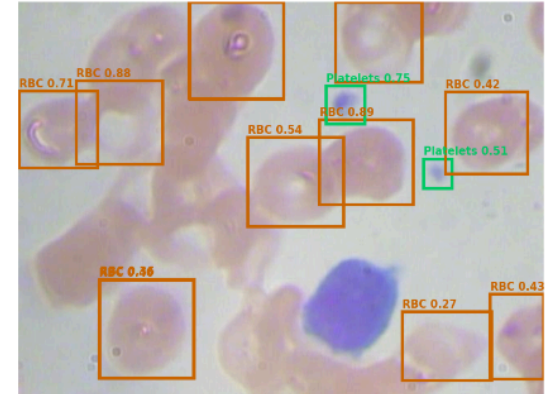
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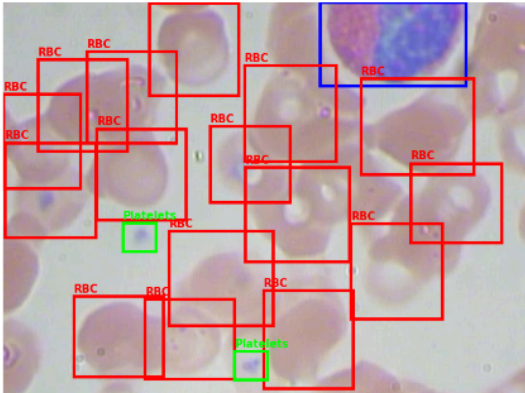
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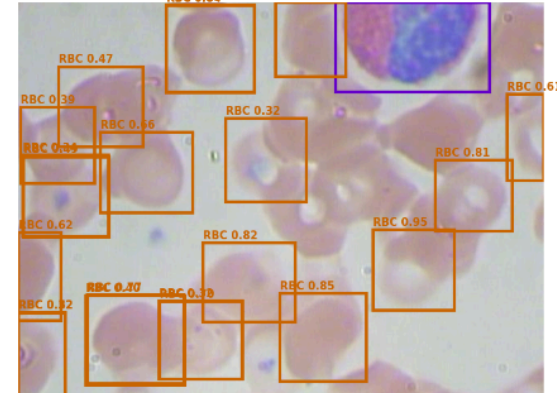
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Pred — BloodImage_00026.jpg.rf.7937b64ebbc36c4e543219b113a85501.jpg



4. Discussion

4.1 Dataset Class Imbalance

The training set is heavily dominated by RBC instances (~85% of all labels). Despite this, WBC, the minority class alongside Platelet, achieved the best AP. This counterintuitive result highlights that class frequency matters less than visual distinctiveness. WBCs are large, uniquely stained, and well-separated from other cells, making them inherently easier to detect. RBC's poor performance despite abundance suggests that density and overlap are the primary challenge, not data quantity.

4.2 Confidence Threshold Analysis

The confidence threshold controls the precision-recall trade-off at inference time. For clinical deployment, threshold selection has direct patient safety implications:

Threshold	Precision / Recall	Clinical Risk	Recommendation
0.25	Low P / High R	High false positives	Development / exploration only
0.55	Balanced	Moderate on both sides	Recommended for clinical screening
0.70+	High P / Low R	Misses real cells (false negatives)	Too conservative for screening

A threshold of 0.55 is recommended as a starting point for clinical screening. This balances false positive reduction (avoiding spurious alerts) with sufficient recall to catch genuine pathology. Per-class threshold tuning, setting a lower threshold for WBC and Platelet (where false negatives are more dangerous) and a higher threshold for RBC, would further optimize clinical performance.

4.3 False Positives vs. False Negatives in Medical Context

In this medical domain, false negatives are more dangerous than false positives. The confusion matrix shows that 202 RBCs and 10 Platelets were missed entirely (classified as background). In a clinical CBC context:

- 202 missed RBCs could underestimate the RBC count and mask an anemia diagnosis, potentially delaying treatment for a patient with iron deficiency or hemolytic disease.
- 10 missed Platelets, while small in absolute number, could contribute to an underestimated platelet count in a patient with thrombocytopenia, where low platelets increase bleeding risk.
- False positives (240 RBC, 28 Platelet) are less dangerous as they primarily result in inflated counts that would prompt re-examination, rather than causing a missed diagnosis.

This asymmetry reinforces the recommendation to use a lower confidence threshold (higher recall) for WBC and Platelet classes, accepting more false positives in exchange for better sensitivity.

4.4 Business Viability and Deployment Recommendation

The model is not yet ready for standalone clinical diagnosis, but is suitable for deployment as a screening aid with mandatory pathologist oversight. The following improvements are recommended before any clinical claim:

1. Expand the training dataset to 2,000–5,000 images from diverse labs and staining protocols.

2. Apply SAHI (Slicing Aided Hyper Inference) or train at 1280px resolution to improve small Platelet detection.
3. Validate on an independent clinical test set from a separate institution.
4. Implement per-class confidence threshold tuning calibrated to clinical stakes.
5. Pursue Health Canada Class II medical device approval before any diagnostic application.

As an immediate deployment path, the model can be used for internal lab quality control and automated cell counting in research settings, where errors are caught by downstream review processes.

5. References

Nelson, J. (2022). BCCD Dataset (Version 3) [Dataset]. Roboflow Universe.
<https://universe.roboflow.com/joseph-nelson/bccd>

Ultralytics. (2026). YOLO26 Documentation. Ultralytics. <https://docs.ultralytics.com>